THE UNIVERSITY OF MELBOURNE

Department of Finance
Faculty of Business and Economics
FNCE 30007 - Derivative Securities
FINAL EXAMINATION: Semester 1, 2022
Exam Duration: Three (3) Hours writing time
15 minutes reading time

Instructions to Candidates

1. This is an OPEN BOOK, OPEN NOTES examination.
2. No formulae sheet is provided.
3. This examination contains **10 QUESTIONS for a total of 75 marks**. You are required to attempt **ALL** questions.
4. Use the Standard Normal Tables provided on Canvas under "Modules/Final Exam Information". You will need to round d1 and d2 in the BSM model to 4 decimal digits. Consequently, you will need to use interpolation on the tables and show your workings.
5. For calculations other than the z-scores, the exam questions will specify if you need to round and, if so, to how any decimal digits. The question will also specify whether the rounding needs to take place at each step or only for the final calculation.
6. The exam requires file uploads. These files may take some time to complete their upload in peak exam times. Students are permitted up to 30 minutes after the scheduled completion time (writing and reading time) to upload and check files. File uploads must be fully completed by this time: for example, students sitting the Derivatives exam at 3:00pm on June 10th with 15 minutes reading time + 180 minutes writing time + 30 minutes late submission window will have a final submission time of 6:45 pm. No penalty will be applied if you upload within the 30 minutes of the late submission window (or, of course, at any other time prior to the start of those 30 minutes). Students who were prevented from submitting on time or at all due to technical difficulties will need to apply for technical/special consideration with supporting documentation. Please notice that the late submission period cannot be used as additional exam time.
7. All answers **MUST BE HANDWRITTEN:** typed answers will be awarded **ZERO marks**.
8. You can hand-write your answers on blank or lightly ruled physical pages or on your iPad, tablet or similar digital device (but you are responsible for any related tech issues).

9. Make sure you are **NOT** overwriting a question with your answers.
10. If you are writing on paper (i.e., physical pages), please scan your answers using a scanner or a mobile device. Detailed instructions about scanning on mobile devices are available via https://lms.unimelb.edu.au/students/student-guides/gradescope-converting-images-to-pdf. All answers submitted **MUST BE LEGIBLE**, illegible (unreadable) answers will be awarded **ZERO** marks.
11. Answers need to be uploaded and assigned to each of the questions.
12. Please, upload **only ONE PDF** containing all your answers, NOT one PDF for each answer.
13. **DO NOT upload images.**
14. From the commencement of the exam (e.g. starting at 3:00 PM AEST on June 10th for students sitting the Derivatives exam on that day) and for the first 45 minutes (e.g. until 3:45 PM AEST on June 10th), teaching staff will be available online to address students' questions about exam content.
15. The above mentioned communication between students and teaching staff will take place through the Big Blue Button tool (or, Exam Support online tool). The tool will be accessed from the subject navigation menu in the LMS. The tool will be visible at the commencement of the exam and for the first 45 minutes. Further:
    a. Students can query exam content to teaching staff and are unable to use their microphones to ask questions.
    b. Students are able to contact teaching staff only: they are unable to connect with other students during an exam.
16. If necessary, any exam correction or clarification to content will be posted as an Announcement on the LMS. Please keep an eye on the Announcements for the subject, either on the LMS or through your email.
17. Once the Exam Support tool is closed (e.g., at 3:45 PM AEST on June 10th for students sitting the Derivatives exam on that day), students are advised to contact 13MELB (13 6352 - Option 1 - Exams) if they have queries; from Outside Australia: +61 3 9035 5511.
18. For any issues other than exam content (including those arising during the first 45 minutes of the exam), students are also recommended to contact 13MELB (13 6352 - Option 1 - Exams); from Outside Australia: +61 3 9035 5511.
19. Recommended **browsers**. The University has recommended **Google Chrome** as a reliable browser due to technical issues with Safari users. **Firefox** has also been shown to be reliable.
20. Additional Student Assistance Information can be found at the following links:

a. Student technical support webpage: https://universityofmelbourne.cmail20.com/t/d-l-ciumc-tykjsuukl-m/

b. Special consideration: https://universityofmelbourne.cmail20.com/t/d-l-ciumc-tykjsuukl-c/
c. Technical consideration: https://universityofmelbourne.cmail20.com/t/d-l-ciumc-tykjsuukl-q/

21. Communication and Collusion between students is absolutely forbidden and will result in very serious consequences.

QUESTION 1 [2+2+2+3+3+2 = 14 MARKS]

In late October 2021, Steven A. Smith, the portfolio manager of the KGD investment fund, predicted a big downward correction in the Australian equity market. He, thus, decided to put the entire portfolio value, AUD 2.5 millions as of 30/10/2021, into short positions on a few dozen stocks listed on the ASX.

Below are some market information and performance statistics since the fund has taken the above mentioned large bearish bets:

	Performance Measures			
	Return	Beta	Alpha	Dividend Yield
KGD Fund	7.40%	-0.91	0.60%	0%
ASX200 Spot	- 9.50%	1	0%	4.10%
ASX200 Futures	- 9.10%	1	0%	N.A
AUS Risk Free rate (current as of 30/04/2022): 1.25%				
ASX200 Spot Level (April 30, 2022 close): 7,287				
Note: - The return and alpha refer to the 30/10/2021- 30/4/2022 period - Dividend Yield and Risk free rate are per annum and continuously compounded				

	Correlation Matrix of Returns – Annualized		
	ASX200 Spot	ASX200 Futures	KGD Fund
ASX200 Spot	1		
ASX200 Futures	0.988	1	
KGD Fund	-0.910	-0.920	1

Despite the actual downward movement in the market since he took the short positions, Steven A. still believes as of today, April 30 2022, that the ASX 200 will suffer further losses over the next two quarters. In facts, he expects the ASX 200 to generate a -5% return (including dividend yield) over the next six months, while he believes his portfolio will deliver the same alpha (0.60% over six months). However, the investment committee (IC) of the KGD fund does not fully agree with him. Upon extensive discussions, the portfolio manager and the IC agree to keep the short positions but also to make the portfolio less sensitive to fluctuations in the overall equity market. Practically, they decide that an overall beta of -0.2 is adequate and that futures contracts on the ASX 200 are the appropriate tool to reach such goal.

The specs for index futures contract are reported below.

Contract Specification for ASX SPI 200™ Futures

CONTRACT	ASX SPI 200™ FUTURES
Commodity	Code AP
Contract Unit	Valued at A$25 per index point (e.g. A$117,500 at 4,700 index points).
Contract Months	March/June/September/December up to six quarter months ahead and the nearest two non-quarterly expiry months
Minimum Price Movement	One index point (A$25)
Exercise Prices	Set at intervals of 25 index points
Last Trading Day[2]	All trading in expiring contracts ceases at 12.00pm on the third Thursday of the settlement month. Non-expiring contracts underlying futures contract will continue to trade as per the trading hours.
Cash Settlement Price[2]	The Special Opening Quotation of the underlying S&P/ASX 200 Index on the Last Trading Day. The Special Opening Quotation is calculated using the first traded price of each component stock in the S&P/ASX 200 Index on the Last Trading Day, irrespective of when those stocks first trade in the ASX trading day. This means that the first traded price of each component stock may occur at any time between ASX market open and ASX market close (including the Closing Single Price Auction) on the Last Trading Day. Should any component stock not have traded by ASX market close on the Last Trading Day, the last traded price of that stock will be used to calculate the Special Opening Quotation.
Trading Hours[2]	5.10pm to 7.00am and 9.50am to 4.30pm (during US daylight saving time) 5.10pm to 8.00am and 9.50am to 4.30pm (during US non daylight saving time)
Settlement Day	The first business day after expiry, ASX Clear (Futures) publishes the final settlement price of the contract. On the second business day after expiry, ASX Clear (Futures) settles cash flows as a result of the settlement price.

2 Unless otherwise indicated, all times are Sydney times. US daylight saving begins first Sunday in April and ends last Sunday in October.

Based on the above information:

a) What futures position should the KGD fund take to achieve the goal stated above? Explain your answer.

Note: you are not required to work out the number of contracts in this part of the question.

b) What specific ASX200 futures contract (i.e., what maturity) should Steven A. choose? Explain your answer.

c) From the KGD fund point of view, the ASX spot and futures markets present essentially no arbitrage opportunities. Assuming that futures contracts expire on the last day of the month, what futures price should Steven A. expect to trade at, if he puts the hedge in place today (30/04/2022) and for the next two quarters?

Show all your detailed workings and round to 4 decimal digits at each step

d) How many futures contracts should Steven A. trade to reach the hedging goal stated above? **Show all your detailed workings and round the final answer to the nearest integer.**

e) With the above hedging strategy in place, what risks (if any) will the KGD fund be exposed to until expiration of the futures position? Explain your answer.

f) The IC asks Steven A. what return the hedged portfolio is expected to earn during the hedging period. Given the hedging strategy you suggest above and given all the information provided, what should Steve A. expect to earn on his overall (i.e., hedged) position during the hedging period? **Show all your detailed workings.**

Note: you can assume that the riskfree rate over the upcoming six-months is half of the annualized rate reported above (i.e, 1.25%/2 = 0.625%)

QUESTION 2 [3 MARKS]

As an arbitrageur in the FX market, you are checking the Australian Dollar (AUD) Indian Rupee (INR) pair. As of today (31 May 2022), you observe a spot exchange rate of 55.91 INR per AUD. In Australia the risk-free rate is now at 3.50%, while in India is at 3.98%. Those interest rates are per annum, with continuous compounding.

You see the following forward rates on your screen:

Maturity	**Rate**
September 2022	55.3505
November 2022	56.0444

Is there an arbitrage profit opportunity? If so, how could you take advantage of it?

Note: Ignore transaction costs and assume the forward contracts expire at the of the respective month.

Show all your detailed workings and round to 4 decimal digits at each step.

QUESTION 3 [6 MARKS]

Celticoin, a heavily traded cryptocurrency, has a current price of $450 in the spot market. Jayson Tatum is offering his friend Marcus Smart the following deal: "If you pay me $3,000 today, in 2 months I will pay you a dollar for each dollar Celticoin's spot price at that point in time is below its spot price now. Otherwise I will pay you nothing. Actually, you know what, if Celticoin in 2 months is below where it is now, I will pay you the square of the difference! With T-bill rates at 2%, this should be a no brainer for you."
Marcus asks whether he can get his payoff at any time before the end of the 2 months in case the price of Celticoin drops significantly during the period. But Jayson replies "No, the deal is that we need to wait until the end of the 2 months and, then, see whether I pay you something or nothing based on what the price of Celticoin is at *that* point"
Marcus, then, calls up his financial advisor, Al Horford, a well-known expert in cryptocurrencies. Horford believes that, in each of the next 2 months Celticoin could indeed go down and by a large amount, but given its high volatility, it may even go up, although not as much. When asked for numbers, Horford says "Look, it may go down 15% every month but I think it could also go up by 10% each month!"

Taking into account Horford's prediction, use the binomial pricing model (specifically a two-step tree) to assess whether the price Jayson proposes is a fair deal from Marcus' point of view. You can assume that Celticoin has zero storage cost and pays no yield.

Note: The T-Bill (i.e., risk-free) rate Jayson refers to is per annum, continuously compounded.

Show all your detailed workings and round to 4 decimal digits at each step.

QUESTION 4 [1+2+1=4 MARKS]

Given the relative lack of interest in silver in recent months, Joe Davola believes that silver price will continue to display low volatility in the coming months. Silver is trading at $35 in the spot market and pays no yield.

a) Given his beliefs, which one of the following strategies makes sense for Mr. Davola?
   Note: Please, choose only one strategy and explain the reasons for your choice
   1. A short (or, reverse) calendar spread
   2. A long strangle
   3. A Short straddle
   4. A Bear spread using Calls

b) To implement the chosen strategy, Mr Davola is considering an at-the-money call option, currently trading for an $8 premium, and an ATM put option with the same expiration as the call, which is available for a $5 premium.

   Construct a table that shows the profit from the strategy you chose in a) above.

c) For what range of stock prices would the strategy you chose in a) lead to a loss?

QUESTION 5 [4 MARKS]

Coco is enrolled at the University of Melbourne and has just completed Principles of Finance where, among other things, she has learnt about the existence of the futures market. Coco has a better understanding of the spot market and she is not quite sure about how investments in the two markets would compare. Building on her knowledge from Principles, Coco would like to invest in the ASX 200, as a broadly diversified portfolio. She approaches her friend Iga and asks her "Should I consider investing in the ASX200 using index futures contracts? Would I get a better or worse return than if I invested through a passive index fund on the ASX?" Iga is also a student at Melbourne Uni and has successfully completed Derivative Securities.
Using her Derivatives expertise, Iga replies: "Well, as long as you hold your, say, long futures position for a relatively short time so that interest rates do not change and the dividend yield does not change, then it doesn't matter which one you choose: the return that you get from your index futures position is going to match the excess return on the index fund over the same time period".

Would you agree with Iga's argument? Why or why not?

NOTE: To answer the question you need to support your assessment with analytics
Hint: Use the no-arbitrage relation between spot price and futures price. Ignore transactions costs and taxes.

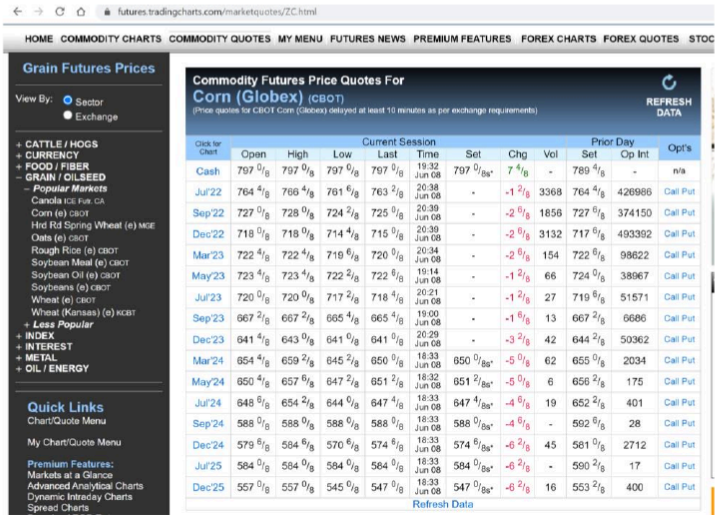
QUESTION 6 [2+2 +1 = 5 MARKS]

Among several other hedging strategies, we have examined delta hedging.
a) How would you characterize delta hedging?
b) Next, explain how the concept is used in the option pricing models we considered in our subject.
c) Finally, illustrate how delta hedging relates to the concept of risk neutral valuation.

QUESTION 7 [1 + 1 + 2 + 1 + 1 + 2+ 2 + 1=11 MARKS]

Jay Peterman has been closely watching the US corn market. Suzie and Elaine, Jay Peterman's investment consultants, have recently estimated that the risk premium in the corn market is negative at 130 cents per bushel per month (with discrete compounding).

Today is June 15th, 2022. While watching his trading screen, Peterman sees the following:



futures.tradingcharts.com/marketquotes/ZC.html

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Grain Futures Prices

View By: Sector / Exchange

+ CATTLE / HOGS
+ CURRENCY
+ FOOD / FIBER
– GRAIN / OILSEED
– *Popular Markets*
Canola ICE Fut. CA
Corn (e) CBOT
Hrd Rd Spring Wheat (e) MGE
Oats (e) CBOT
Rough Rice (e) CBOT
Soybean Meal (e) CBOT
Soybean Oil (e) CBOT
Soybeans (e) CBOT
Wheat (e) CBOT
Wheat (Kansas) (e) KCBT
+ *Less Popular*
+ INDEX
+ INTEREST
+ METAL
+ OIL / ENERGY

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Advanced Analytical Charts
Dynamic Intraday Charts
Spread Charts

Commodity Futures Price Quotes For
Corn (Globex) (CBOT)
(Price quotes for CBOT Corn (Globex) delayed at least 10 minutes as per exchange requirements)
REFRESH DATA

Click for Chart	Current Session								Prior Day		Opt's
	Open	High	Low	Last	Time	Set	Chg	Vol	Set	Op Int	
Cash	797 0/8	797 0/8	797 0/8	797 0/8	19:32 Jun 08	797 0/8a	7 4/8	-	789 4/8	-	n/a
Jul'22	764 4/8	766 4/8	761 6/8	763 2/8	20:38 Jun 08	-	-1 2/8	3368	764 4/8	426986	Call Put
Sep'22	727 0/8	728 0/8	724 2/8	725 0/8	20:39 Jun 08	-	-2 0/8	1856	727 6/8	374150	Call Put
Dec'22	718 0/8	718 0/8	714 4/8	715 0/8	20:39 Jun 08	-	-2 6/8	3132	717 6/8	493392	Call Put
Mar'23	722 4/8	722 4/8	719 6/8	720 0/8	20:34 Jun 08	-	-2 6/8	154	722 6/8	98622	Call Put
May'23	723 4/8	723 4/8	722 2/8	722 6/8	19:14 Jun 08	-	-1 2/8	66	724 0/8	38967	Call Put
Jul'23	720 0/8	720 0/8	717 2/8	718 4/8	20:21 Jun 08	-	-1 2/8	27	719 6/8	51571	Call Put
Sep'23	667 2/8	667 2/8	665 4/8	665 4/8	19:00 Jun 08	-	-1 6/8	13	667 2/8	6686	Call Put
Dec'23	641 4/8	643 0/8	641 0/8	641 0/8	20:29 Jun 08	-	-3 2/8	42	644 2/8	50362	Call Put
Mar'24	654 4/8	659 2/8	645 2/8	650 0/8	18:33 Jun 08	650 0/8a	-5 0/8	62	655 0/8	2034	Call Put
May'24	650 4/8	657 6/8	647 2/8	651 2/8	18:32 Jun 08	651 2/8a	-5 0/8	6	656 2/8	175	Call Put
Jul'24	648 6/8	654 2/8	644 0/8	647 4/8	18:33 Jun 08	647 4/8a	-4 6/8	19	652 2/8	401	Call Put
Sep'24	588 0/8	588 0/8	588 0/8	588 0/8	18:33 Jun 08	588 0/8a	-4 6/8	-	592 6/8	28	Call Put
Dec'24	579 6/8	584 6/8	570 6/8	574 6/8	18:33 Jun 08	574 6/8a	-6 2/8	45	581 0/8	2712	Call Put
Jul'25	584 0/8	584 0/8	584 0/8	584 0/8	18:33 Jun 08	584 0/8a	-6 2/8	-	590 2/8	17	Call Put
Dec'25	557 0/8	557 0/8	545 0/8	547 0/8	18:33 Jun 08	547 0/8a	-6 2/8	16	553 2/8	400	Call Put

Refresh Data

Note 1: The decimal part of the price is quoted in eights. You'll need to translate it into cents. For instance, 724 2/8 means 724 + 2/8 = 724.25. Prices are per bushel.

Note 2: Assume that one can trade at the last recorded prices reported above.

Note 3: Show all your detailed workings and round to 4 decimal digits at each step for all calculations.

a) Is the Corn futures curve upward sloping or downward sloping? Explain your answer

b) Is Corn in Normal Contango or in Normal Backwardation? Justify your answer.

c) Peterman would like to exploit the shape of the futures curve to capture a roll yield. Detail how he might do so in the corn market characterized by the above quotes.

d) Is the roll yield you compute in c) above guaranteed? Why or why not?

e) Based on the information above, what is the spot price of corn expected today for September (i.e., three months from today)?

f) Assume the September contract expires exactly in three months and the risk-free rate of interest is 2% per year with discrete compounding and 1.9803% per year with continuous compounding. The storage fee for corn is 35 cents per bushel per quarter, payable at the end of each quarter. Focusing on the September futures, what annualized continuously compounded convenience yield, if any, is the market attributing to corn?

g) Peterman is considering corn futures for hedging purposes as well. According to the hedging pressure hypothesis, should Peterman expect to earn the 130 cents per bushel risk premium? Why or why not?

h) How would you interpret the current futures price of the July contract if there was no risk premium for corn?

QUESTION 8 [5+2+2 = 9 MARKS]

Steve Rogers owns 2,000 shares of CBA stock, listed on the ASX. Today is June 10, 2022 and CBA is currently trading at $105 per share. Because of the current drop in the banking sector, Steve is worried about further declines over the next 6 months. Steve is, thus, considering trading American put options on CBA with a strike of $100 to protect his position.
As a financial intermediary, you are willing to write the options to Steve. You expect the volatility of CBA to be 8% per month. Also, CBA is expected to pay a dividend of $2 in 3 months from today and a dividend of $1.8 in 9 months from today. With the market expectation of increasing cash rates, you expect the continuously compounded risk-free rate to increase from 0.5% p.a. to 1.5% p.a. in 4 months from today, and, then, stay at that level until the end of the year. Assume each option is written on one share of CBA stock.

Note:
Round u and d to 6 decimal digits.
Round the number of securities to the nearest integer.
For all other calculations, round to 4 decimal digits.
Please show all your detailed workings.

a) Use a three-step binomial tree to calculate the option price.
b) What do you need to do in order to delta hedge your position at the time the options are written? Be specific about the necessary hedging strategy.
c) If CBA stock price increases in the two months after the options are written, how would you keep your position delta hedged? Be specific about the necessary hedging strategy.

QUESTION 9 [4+3 = 7 MARKS]

After seeing the recent huge drop in the share market, you expect the market to recover. You, therefore, traded futures on the ASX200 to speculate on your view. Specifically, you entered into ten September 2022 ASX200 futures contracts on 6th June 2022 when the futures price was 7,175 points. On 8th June 2022, you increased your exposure by entering into another eight September 2022 ASX200 futures contracts at the futures price of 7,065 points. Assume the initial margin required is $7,000 per contract and the maintenance margin is $3,500 per contract. Each index point is worth $50. You are provided with the following futures prices

Date	**Opening price**	**Last price**	**Settlement price**
6th June 2022	7,145	7,080	7,075
7th June 2022	7,073	7,050	7,045
8th June 2022	7,040	7,050	7,052
9th June 2022	7,050	7,060	7,062

Please note: although you have taken two positions (one on June 6th, one on June 8th), they are taken on the same September 2022 contract. As a result, the cash flow from one position may offset the cash flow from the other. You need to look at the overall margin account when answering the questions below.

a) What is your margin account balance at the end of 8th June 2022**? Please show all your detailed workings.**
b) Under what circumstances would you be able to withdraw a maximum amount of $40,000 from the margin account on 10th June 2022. **Please show all your detailed workings.**

QUESTION 10 [1+5+3+3 = 12 MARKS]

Robert Bruce Banner has a portfolio of options on Rio Tinto stocks, as shown in the table below:

Option	Position	Greeks of each option				
		Delta	Gamma	Vega	Rho	Theta
Put	500	-0.509	0.012	44.342	-76.326	-0.025
Call	1000	0.448	0.012	44.586	44.737	-0.006
Put	-600	-0.623	0.011	40.604	-96.965	-0.026

Currently Rio Tinto is trading at $120 per share, the continuously compounded risk-free rate is 1% p.a., the continuously compounded dividend yield on Rio Tinto is 7% p.a. and the volatility of Rio Tinto is expected to 25% p.a. After a detailed analysis of the market, Robert decides to decrease the delta of his portfolio by 500 using a 6-month call option on Rio Tinto with a strike of $115. Delta, Gamma, Vega, Rho and Theta of this 6-month call option are 0.5440, 0.0179, 32.2744, 28.2061 and -0.0111 respectively.

Assume all options are European style, each option is written on 1 share of Rio Tinto stock and Theta is calculated based on calendar days.

Note:

For all answers, please show all your detailed workings.

For underlying asset units and number of options contracts, round to the nearest integer.

For all other calculations, including d1 and d2, round to 4 decimal digits.

a) What view on Rio Tinto stock would justify Robert's decision to decrease the delta of his portfolio?

b) Compute the Delta, Gamma, Vega, Rho and Theta (using calendar days) of Robert's overall option position after trading the additional 6-month call options. Then, explain *in plain English* the meaning of each computed Greek letters.

c) After trading the 6-month call options, Robert shorts 67 shares of Rio Tinto to make his portfolio delta neutral. If the share price of Rio Tinto decreased to $119 instantly, what should Robert do to keep his portfolio delta neutral? What are the risks of a delta neutral position?

d) Assume now that the stock price of Rio Tinto is still $120. Robert is concerned about the risk of a changing risk free interest rate after making his portfolio delta neutral (by shorting 67 shares of Rio Tinto). Robert, hence, wants to hedge his portfolio against this interest rate risk while keeping the portfolio delta neutral. To this end, Robert is considering to trade 9-month put options on the same stock and with a strike of $130. Detail the trades that need to be made in order to achieve those hedging goals. Again, assume each option is written on 1 share of Rio Tinto stock.

----------END OF THE EXAM----------